

the Cloud Security Alliance (CSA). The CSA's Trusted Cloud Initiative programme was created to help cloud service providers to meet industry-recommended standards, offer secure access, ensure compliance management and follow best practices.

Cloud risks can be divided into four major categories:

- Policy and organisational risks
- Technical risks
- Legal risks
- Other risks

Let's examine each of these.

Policy and organisational risks

Lock-in: The lock-in problem takes place when applications, data and services are dependent on a single cloud provider. Being locked in is one of the biggest problems in the cloud computing environment. There could be SaaS lock-in, PaaS lock-in and IaaS lock-in. This problem occurs due to the high level of customisation to meet user needs.

Loss of governance: The cloud provider may subcontract or outsource services to third parties (unknown providers) who may not offer the same guarantees (such as providing the service in a lawful way) that are given by the cloud provider. Or there is a chance that control and operational management of the cloud provider may get changed, so the terms and conditions of their services may also change.

The loss of control and governance could lead to it becoming impossible to fulfill the security requirements. It could lead to a lack of confidentiality, integrity and availability of data, and deterioration in performance and quality of service.

Compliance challenges: Cloud providers make large investments to get external certifications like SAS 70, PCI DSS, HIPPA, etc. In the market, these certifications give them the reputation of following the best security practices. In some cases, a particular certification may also be a problem while accessing cloud services, e.g., if you are using the AWS (Amazon Web Services) cloud and you want to use its EC2 service but the latter doesn't have PCI compliance, then the EC2 service can not be used for credit card related transactions. Although customers can use other services of AWS, they are restrained from this particular service.

Cloud service termination or failure: There must be 24x7 support and high availability of all services. In the competitive world of IT, an inadequate business strategy, lack of financial support and other factors could lead some providers to go out of business or shut down their service portfolio offerings, and it is possible that for a short or medium period of time, some cloud computing services could be terminated.

Supply chain failure: There is the possibility that cloud providers could outsource some services to third parties. In which case, any interruption or corruption in the chain

or a lack of coordination of responsibilities between all the parties involved can lead to the following: inaccessibility of services; loss of data confidentiality, availability and integrity; economic and reputational losses due to failure to meet customer demands; cascading service failure; violation of SLA, etc.

Technical risks

Isolation failure: Since cloud computing works on a multi-tenancy and shared resources architecture, there is a sharing of computing capacity, storage and network among multiple users.

This also leads to some threats like failure of logistics, memory, storage, routing tables (like side channel attacks); SQL injection attacks and guest-hopping attacks where the data of multiple customers can be maliciously stored in the same table.

Resource exhaustion: A cloud service is a fully on-demand, pay-per-use service. There are challenges in the proper allocation of resources to cloud users. Although there are many resource allocation algorithms used for allocating all the resources of cloud services, insufficient resource provisioning and investments in infrastructure may lead to a problem linked to service unavailability or a degradation in performance.

Malicious insiders at cloud service provider firms: The malicious actions of an insider could possibly have an impact on the following: the confidentiality, integrity and availability of all kinds of data, on IP, on all kinds of services and, therefore, indirectly on the organisation's reputation, customer expectations and the experiences of employees.

Taking care of this issue is extremely important because cloud architectures contain certain characteristics that are very high-risk.

Intercepting data in transit: Cloud services are based on a distributed architecture. Therefore, transmission of data takes place across multiple physical machines, from one virtual machine to another, and image distribution is between the cloud infrastructure and remote Web clients, VPN environment, etc.

The vulnerability is more when data is being transferred from on premise to the cloud or vice-versa. Spoofing, man-in-the-middle attacks and sniffing attacks could be possible during data transfer related activities.

Insecure or ineffective deletion of data: Whenever there are alterations at the cloud services provider's end, such as resources being scaled down, physical hardware being moved, etc, data may become available beyond the lifetime specified in the security policy. It may be tough to carry out procedures mandated by the security policy, since full data deletion is only feasible by destroying a disk that also stores data from other clients. When a request to delete a cloud resource is made, this may not result in truly wiping out all the data (as with most operating systems).